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1-Month Internship — Data Science

A practical assignment handbook built around three progressive, portfolio-oriented industry projects.

Internship Domain	Data Science	**Duration**	1 month / 4 weeks
Difficulty	Intermediate	**Projects**	Exactly 3
Who Can Participate	Students, recent graduates, career switchers and interns able to meet the stated prerequisites.	**Working Mode**	Individual, mentor-reviewed, evidence-based delivery.

Learning Objectives

- Apply data science knowledge to a realistic stakeholder problem rather than a tutorial-only exercise.
- Plan work, manage scope, document decisions and communicate progress professionally.
- Produce evidence of quality through testing, validation, review, measurement or structured feedback.
- Finish with artifacts that can be demonstrated in an interview, portfolio or mentor review.

Industry Relevance & Expected Outcome

The internship simulates the expectations of a junior professional working in data science. The intern is expected to move from requirements and planning to execution, validation and final handoff. The three projects intentionally increase in independence and portfolio value.

Recommended Prerequisites & Tools

- Basic programming/domain fundamentals and ability to work independently with documentation.
- Git, documentation tools and the project-specific development/analysis stack.

Month-at-a-Glance

Week 1	Research • requirements • scope • setup • design/method
Week 2	Core execution • development • analysis • creation
Week 3	Advanced work • testing • refinement • validation
Week 4	Finalization • documentation • presentation • submission

General Submission Rules

- Use only authorized/public/simulated data and systems. Never test real systems without explicit written authorization.
- Keep a dated work log and maintain organized project files with clear filenames.
- Cite external sources where research or public data is used; distinguish assumptions from observed evidence.
- Submit a concise README or equivalent handoff document so a mentor can reproduce or review the work.
- Final presentations should demonstrate the result, evidence, key decisions, limitations and next steps.

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Project 1: Customer Churn Investigation

Project Type: Foundation / Core Industry Task

1. Real-World Scenario

Analyze a simulated subscription dataset to identify churn drivers, segment customers and propose retention actions.

2. Project Objective

Deliver a working, reviewable solution for the scenario using professional data science practices. The objective is not only to produce an output, but to show how you moved from problem definition to a defensible final result.

3. Business Use Case

A real organization could use this type of workflow to improve service quality, operational visibility, productivity, decision-making or customer/user experience. The intern should identify the primary stakeholder, the decision enabled by the output and the measurable benefit.

4. Problem Statement

Design and complete the data science project so that a stakeholder can reliably use or evaluate the final result. Define inputs, outputs, constraints, quality measures and the conditions under which the solution should be considered successful.

5. Required Features / Activities

- Data dictionary
- Data cleaning and quality checks
- EDA with business questions
- Evidence-based insights
- Limitations and next-step recommendations

6. Functional / Professional Requirements

- Define the target user/stakeholder, scope boundaries and success metrics before execution.
- Use a repeatable workflow with traceable decisions and clearly organized artifacts.
- Handle realistic edge cases, invalid inputs, exceptions or uncertainty relevant to the domain.
- Include a validation/review step and retain evidence of the result.
- Document limitations and recommend realistic next improvements.

7. Technical / Method Requirements

- Recommended stack: Python, pandas, NumPy, Jupyter, matplotlib/seaborn
- Use current stable tooling that is accessible to the intern; reasonable alternatives are allowed when documented.
- Apply version control or a dated evidence folder appropriate to the domain.
- Follow responsible-use, security, privacy and data-integrity practices relevant to the project.

8. Suggested Technology Stack

Python, pandas, NumPy, Jupyter, matplotlib/seaborn

9. Step-by-Step Implementation Plan

1. Research the data science workflow and identify the target user, stakeholders, inputs, outputs and success metrics.
2. Convert the scenario into a written requirements document, acceptance criteria and delivery checklist.
3. Create the initial architecture/design/data model and establish a version-controlled workspace.
4. Implement the core workflow in small, testable increments and keep a decision log.
5. Validate the result using representative test cases or evidence and document known limitations.

10. Week-by-Week Plan

Week 1	Research, stakeholder/problem framing, requirements, dataset/assets, environment setup and initial design.

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Week 2	Core implementation, analysis, content creation, research execution or prototype build.
Week 3	Advanced requirements, testing/validation, optimization, refinement and mentor review.
Week 4	Final QA/validation, documentation, presentation/demo, packaging and submission.

11. Deliverables

- Project source files or working analysis/design artifacts
- Requirements and implementation/research documentation
- Test/validation evidence
- README or user/technical guide
- Final presentation or demo
- Version-controlled project repository or organized submission folder

12. Expected Final Output

A complete, organized and presentation-ready project package that demonstrates the required workflow, includes evidence of quality/validation, and can be understood by a mentor without relying on undocumented steps.

13. Acceptance Criteria

- All stated core requirements are completed and mapped to evidence.
- The solution/workflow is reproducible by a mentor using the submitted instructions and files.
- Quality checks/tests/validation are documented and major known issues are disclosed.
- The final output is professionally organized, named consistently and ready for demonstration.
- The intern can explain key decisions, trade-offs, limitations and future improvements.

14. Evaluation Rubric — 100 Points

Criterion	Points
Requirements & domain understanding	15
Technical / professional execution	25
Quality, accuracy & validation	20
Problem-solving & independence	15
Documentation & reproducibility	10
Presentation & communication	10
Creativity / improvement thinking	5

15. Portfolio Value

Present the project on a resume as a measurable internship project; publish appropriate source/artifacts on GitHub or a portfolio; share a concise outcome-focused LinkedIn post; and use the project as an interview story covering problem, action, evidence and result.

16. Viva / Interview Questions

- What business or user problem does the project solve, and how did you define success?
- Which design, technical, analytical or research decision had the biggest impact on the result?
- How did you validate the quality or correctness of your work?
- What edge cases, limitations or risks did you identify?
- What would you change if this solution had to support 10× the users/data/workload?
- How would you explain the project to a non-technical stakeholder?
- Which part of the project required the most independent problem-solving?

17. Extension / Advanced Challenge

- Add monitoring/measurement or a stronger evaluation framework and compare results before vs. after.

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- Introduce an additional user/stakeholder role or more realistic edge cases.
- Automate one repetitive step and document the productivity or quality benefit.
- Create a production/research-readiness roadmap covering security, scalability, governance or maintainability.

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Project 2: Retail Demand & Basket Analysis

Project Type: Intermediate Real-World Task

1. Real-World Scenario

Study historical retail transactions to identify demand patterns, product associations and business opportunities.

2. Project Objective

Deliver a working, reviewable solution for the scenario using professional data science practices. The objective is not only to produce an output, but to show how you moved from problem definition to a defensible final result.

3. Business Use Case

A real organization could use this type of workflow to improve service quality, operational visibility, productivity, decision-making or customer/user experience. The intern should identify the primary stakeholder, the decision enabled by the output and the measurable benefit.

4. Problem Statement

Design and complete the data science project so that a stakeholder can reliably use or evaluate the final result. Define inputs, outputs, constraints, quality measures and the conditions under which the solution should be considered successful.

5. Required Features / Activities

- Data dictionary
- Data cleaning and quality checks
- EDA with business questions
- Evidence-based insights
- Limitations and next-step recommendations

6. Functional / Professional Requirements

- Define the target user/stakeholder, scope boundaries and success metrics before execution.
- Use a repeatable workflow with traceable decisions and clearly organized artifacts.
- Handle realistic edge cases, invalid inputs, exceptions or uncertainty relevant to the domain.
- Include a validation/review step and retain evidence of the result.
- Document limitations and recommend realistic next improvements.

7. Technical / Method Requirements

- Recommended stack: Python, pandas, visualization, association analysis
- Use current stable tooling that is accessible to the intern; reasonable alternatives are allowed when documented.
- Apply version control or a dated evidence folder appropriate to the domain.
- Follow responsible-use, security, privacy and data-integrity practices relevant to the project.

8. Suggested Technology Stack

Python, pandas, visualization, association analysis

9. Step-by-Step Implementation Plan

1. Independently refine requirements after reviewing edge cases, constraints and stakeholder priorities.
2. Build the main solution with a clear separation between inputs, processing, outputs and reporting.
3. Add validation, error handling, quality controls and measurable performance/accuracy/usability checks.
4. Run a structured review against the acceptance criteria and correct the highest-impact gaps.
5. Prepare a concise professional demonstration supported by evidence and documentation.

10. Week-by-Week Plan

Week 1	Research, stakeholder/problem framing, requirements, dataset/assets, environment setup and initial design.

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Week 2	Core implementation, analysis, content creation, research execution or prototype build.
Week 3	Advanced requirements, testing/validation, optimization, refinement and mentor review.
Week 4	Final QA/validation, documentation, presentation/demo, packaging and submission.

11. Deliverables

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Project 3: Workforce Analytics Study

Project Type: Capstone / Portfolio Task

1. Real-World Scenario

Analyze workforce data for hiring, attrition and productivity indicators while documenting limitations and responsible interpretation.

2. Project Objective

Deliver a working, reviewable solution for the scenario using professional data science practices. The objective is not only to produce an output, but to show how you moved from problem definition to a defensible final result.

3. Business Use Case

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4. Problem Statement

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5. Required Features / Activities

- Data dictionary
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6. Functional / Professional Requirements

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- Use a repeatable workflow with traceable decisions and clearly organized artifacts.
- Handle realistic edge cases, invalid inputs, exceptions or uncertainty relevant to the domain.
- Include a validation/review step and retain evidence of the result.
- Document limitations and recommend realistic next improvements.

7. Technical / Method Requirements

- Recommended stack: Python, pandas, visualization, statistical analysis
- Use current stable tooling that is accessible to the intern; reasonable alternatives are allowed when documented.
- Apply version control or a dated evidence folder appropriate to the domain.
- Follow responsible-use, security, privacy and data-integrity practices relevant to the project.

8. Suggested Technology Stack

Python, pandas, visualization, statistical analysis

9. Step-by-Step Implementation Plan

1. Frame the capstone as a portfolio project with a defined business owner, user group and measurable outcome.
2. Design the end-to-end solution, including architecture/design/research method, data or assets and risk controls.
3. Implement the highest-value features and perform testing, validation, optimization or peer review.
4. Package the work as a professional handoff with reproducibility, documentation and presentation materials.
5. Reflect on trade-offs, limitations and realistic next steps for a production or research environment.

10. Week-by-Week Plan

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